The Convex Origin of Fixed Costs[†]

Hanbaek Lee[‡]

University of Cambridge

March 16, 2026

Abstract

Are fixed and convex adjustment costs different theories, or the same economics? This paper proves the latter. When fixed costs are heterogeneous — drawn from a distribution G — the optimal threshold rule is exactly a convex cost model: same payoffs, same decisions, same sample paths. The deterministic fixed cost is the singular exception; any heterogeneity restores convexity. Micro behaviour is always lumpy; at the aggregate level, there is no type distinction between lumpy and smooth — only a degree distinction governed by G. Applied to pricing, G parametrises the Calvo–Golosov-Lucas spectrum and decomposes non-neutrality into frequency and selection. G unifies macroeconomic frictions, discrete choice, rational inattention, and control costs.

Keywords: Fixed costs, convex adjustment costs, Fenchel duality, menu costs, Calvo pricing, Phillips curve.

JEL codes: E31, E32, D81.

[†]I am grateful for helpful comments and discussions from Philip Ushchev. All errors are my own.
[‡]Email: hl610@cam.ac.uk

1 Introduction

Macroeconomic models of adjustment frictions come in two seemingly distinct forms: fixed costs that produce binary, "lumpy" adjustment, and convex costs that produce smooth, gradual responses. This paper proves they are the same. When fixed costs are heterogeneous — firms draw their cost from a distribution G — the optimal threshold rule is exactly a convex cost model. The equivalence is not an approximation: it delivers the same payoffs, the same adjustment probabilities, and the same realised binary outcomes. The reason is simple: heterogeneous cost draws make each additional adjuster more expensive than the last, so the aggregate cost of adjustment is convex — precisely the Fenchel conjugate of G.

The paper makes three contributions. First, it establishes the duality and its structural implications. The deterministic fixed cost — where every firm faces the same cost — is a knife-edge singularity; any heterogeneity, however small, restores strict convexity (Proposition 2). At the micro level, each firm's decision is always binary. At the aggregate level, there is no type distinction between "lumpy" and "smooth" — only a degree distinction, with the shape of G as the dial. Uniform G gives textbook quadratic costs (Khan and Thomas, 2008); concentrated G gives the Ss rule of Golosov and Lucas (2007); dispersed G approximates Calvo (1983). These are not different kinds of models but different shapes of one distribution. This also provides a partial-equilibrium foundation for the irrelevance result of Khan and Thomas (2008) (see also Gourio and Kashyap 2007): with uniform fixed costs, each firm's problem is already convex — a representational artifact, not an economic force to be averaged out.

Second, G acts as a Rosetta Stone across literatures. Different distributions generate cost functions corresponding to well-known information-theoretic entropies: exponential G yields Shannon entropy, the cost function in the binary rational inattention model of Matějka and McKay (2015); uniform G yields the quadratic attention penalty of Gabaix (2014); power law G yields Tsallis q-entropy (Tsallis, 1988); Gumbel-distributed costs recover the multinomial logit of McFadden (1974). Five literatures — macroeconomic frictions, discrete choice, rational inattention, sparse decision-making, and control costs (Mattsson and Weibull, 2002; Hofbauer and Sandholm, 2002) — are manifestations of one duality under different distributional assumptions on G (Proposition 4). These are correspondences of functional form; the

economic content and welfare implications of each literature may differ.

Third, it develops a pricing application that bridges Calvo and Golosov–Lucas. The menu cost distribution G is the structural primitive: the repricing hazard $\Lambda(x) = G(Bx^2/w)$ determines who adjusts, and its shape governs how strongly adjustment is selected toward high-gap firms (Definition 1). A power law $G(\xi) = (\xi/\bar{\xi})^\gamma$ spans this spectrum continuously through γ (Proposition 13).[1] The resulting state-dependent Phillips curve (Proposition 11) decomposes into frequency — how often firms reprice — and selection — how concentrated adjustments are among high-gap firms. Two economies with identical repricing frequency can differ dramatically in non-neutrality; the difference is the *selection wedge* $\kappa - \bar{\Lambda}$, determined entirely by G (Proposition 12). This complements Alvarez et al. (2016): their observable moments — kurtosis and frequency — are endogenous projections of G, so G provides the structural layer from which their empirical diagnostics are derived.

Although the pricing application is developed in greatest depth, the duality applies to any extensive-margin decision with heterogeneous costs, including firm entry and exit, technology adoption, default, labour force participation, search and matching, and human capital investment. In each case, the distribution of the friction — not the label "fixed" or "convex" — is the economically meaningful object.

Related literature The equivalence connects five literatures. First, the macroeconomic fixed cost tradition: Caplin and Spulber (1987) on monetary neutrality under uniform price gaps; Gourio and Kashyap (2007), Khan and Thomas (2008), and Oskolkov and Lippi (2026) on investment; Dotsey et al. (1999), Golosov and Lucas (2007), Midrigan (2011), and Nakamura and Steinsson (2010) on pricing; Alvarez et al. (2016) on sufficient statistics for non-neutrality; Vavra (2014) on the cross-sectional variance of price changes as a predictor of non-neutrality. In the present framework, the Vavra (2014) sufficient statistic is an endogenous function of G, so G governs both the variance and the structural friction generating it. The paper also contributes to the generalised Ss literature of Caballero and Engel (1993, 1999). Their framework introduces an "adjustment function" — the probability of adjustment as a function of the agent's state — as the primitive governing aggregate dynamics. The repricing hazard $\Lambda(x)$ in Definition 1 is the adjustment function for this class of

[1] The framework does not literally nest Calvo, since $\Lambda(x) = G(Bx^2/w)$ is always increasing in $|x|$ and hence never exactly constant. The approximation is made precise in Section 4.

models; its shape is not an assumption but a derived object, determined entirely by the menu cost distribution G. The present paper thus provides a micro-foundation for the adjustment function approach: G is the structural primitive from which the adjustment function is identified. Caballero and Engel (2007) show that the variance of idiosyncratic shocks relative to the inaction band governs whether state dependence amplifies or dampens aggregate responses; Proposition 13 makes an analogous monotonicity statement in terms of γ. Second, discrete choice theory: the Williams-Daly-Zachary theorem (McFadden, 1974, 1978, 1981) shows that the social surplus function is the convex conjugate of a generating function determined by the shock distribution; Theorem 1 is the binary-choice, cost-heterogeneity analogue. Third, control costs (Mattsson and Weibull, 2002; Hofbauer and Sandholm, 2002): the fixed cost model *is* a control cost model, with the penalty endogenously determined by G. Fourth, rational inattention (Sims, 2003; Matějka and McKay, 2015): Shannon entropy costs are equivalent to exponentially-distributed fixed costs for binary choice. Fifth, sparse decision-making (Gabaix, 2014): quadratic attention costs are equivalent to uniformly-distributed fixed costs. All five are manifestations of a single duality under different distributional assumptions on G.

Outline Section 2 presents the duality theorem, characterises the induced cost function, establishes the universality of the convex representation, and develops the entropy correspondence. Section 3 applies the result to investment. Section 4 develops the menu cost–Calvo spectrum, including the repricing hazard, the selection effect, the state-dependent Phillips curve, and the non-neutrality decomposition. Section 5 discusses further applications. Section 6 discusses broader implications. Section 7 concludes.

2 The duality

2.1 Primitives

Consider an agent who faces a binary decision: take an action ("adjust") or maintain the status quo ("do nothing"). Adjusting yields a gross surplus $S \geq 0$ but requires paying a fixed cost $w\xi$, where $w > 0$ is a wage or unit cost and ξ is a random draw from a distribution G on $[0, \bar{\xi}]$ with density $g > 0$. Not adjusting yields zero. The

randomness in ξ captures heterogeneity in the cost of adjustment. This heterogeneity may be across agents (a cross-section of firms facing different menu costs) or across time (a single firm drawing a new cost each period, as in Dotsey et al. 1999).

This setup is minimal. It strips away the details of particular applications to isolate the structure common to all of them. In the investment application of Section 3, S is the surplus from investing at the optimal scale. In the pricing application of Section 4, S is the quadratic gain from closing the price gap. Because the setup is general, the duality theorem applies to any extensive-margin decision where the agent's cost of acting is uncertain.

Fixed cost formulation The agent observes the realised cost ξ and then decides whether to adjust. Adjustment is optimal if and only if the surplus exceeds the cost: $w\xi \leq S$. The expected payoff, taken over the distribution of cost realisations, is

$$\Pi^{FC}(S) = \int_0^{\bar{\xi}} \max\{S - w\xi,\ 0\}\, dG(\xi) = \int_0^{\min\{S/w,\, \bar{\xi}\}} (S - w\xi)\, dG(\xi). \tag{1}$$

The agent adjusts for all cost draws below the threshold $\xi^* = S/w$ and does not adjust otherwise. This is the standard setup in the fixed cost literature.

Probability choice formulation In an alternative formulation, the agent does not observe ξ directly but instead chooses an adjustment probability $\varphi \in [0, 1]$, paying a cost $C(\varphi)$ that is increasing and convex in φ:

$$\Pi^{PC}(S) = \max_{\varphi \in [0,1]} \big\{\varphi \cdot S - C(\varphi)\big\}, \tag{2}$$

where $C : [0, 1] \to \mathbb{R}_+$ with $C(0) = 0$. The agent trades off the expected benefit of adjusting ($\varphi \cdot S$) against the cost of choosing a high adjustment probability ($C(\varphi)$). This is the structure of control cost models in game theory and of Calvo-like pricing models where the repricing probability is a primitive.

The duality theorem establishes that these two formulations are not merely analogous but identical. For each distribution G with a positive density, there exists a unique convex cost function C such that the two models yield the same payoff and the same optimal adjustment probability.

2.2 Main result

Theorem 1 (Convex duality of fixed costs)**.**
The following are equivalent:

(i) The agent uses the threshold rule under G, obtaining $\Pi^{FC}(S)$.

(ii) The agent solves the probability choice problem (2) with cost function

$$C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi \, dG(\xi). \tag{3}$$

The two formulations yield the same payoff $\Pi^{FC}(S) = \Pi^{PC}(S)$ for all $S \geq 0$ and the same optimal adjustment probability $\varphi^* = G(S/w)$. Moreover, there exists a coupling under which the realised adjustment indicators coincide almost surely. If the probability choice model randomises by drawing $\xi \sim G$ and adjusting whenever $\xi \leq G^{-1}(\varphi^*)$, the two models produce identical binary outcomes path by path.

Moreover, $\Pi^{PC}(S) = C^*(S)$, the Fenchel conjugate of C, and $C(\varphi) = (\Pi^{FC})^*(\varphi)$, so the two formulations are conjugate duals.

Proof.
Substituting the threshold $\xi^* = S/w$ (assuming $S/w \leq \bar{\xi}$; the corner is analogous) into (1):

$$\Pi^{FC}(S) = S \cdot G(S/w) - w \int_0^{S/w} \xi \, dG(\xi). \tag{4}$$

The first term is the expected surplus from adjusting: the probability of adjustment $G(S/w)$ times the surplus S. The second term is the expected cost paid by adjusters: the average cost among agents whose draw falls below the threshold. Setting $\varphi = G(S/w)$ so that $S/w = G^{-1}(\varphi)$:

$$\Pi^{FC}(S) = \varphi \cdot S - w \int_0^{G^{-1}(\varphi)} \xi \, dG(\xi) = \varphi S - C(\varphi).$$

This is the objective of (2) evaluated at $\varphi^* = G(S/w)$. To confirm that this φ^* is indeed optimal for the probability choice problem, note that the first-order condition of (2) is $S = C'(\varphi)$. Since $C'(\varphi) = wG^{-1}(\varphi)$ (Proposition 1 below), this gives $G^{-1}(\varphi^*) = S/w$, i.e., $\varphi^* = G(S/w)$ — the same threshold probability. The conjugate

relationship follows from the definition of the Fenchel conjugate: $C^*(S) = \sup_\varphi\{\varphi S - C(\varphi)\} = \Pi^{PC}(S)$.

For the coupling: in the fixed cost model, the agent draws $\xi \sim G$ and adjusts whenever $\xi \leq \xi^* = S/w$. This occurs with probability $G(\xi^*) = \varphi^*$. The probability choice model specifies only that adjustment occurs with probability φ^*. The particular randomisation device is not part of the formulation. However, one natural implementation is to draw $\xi \sim G$ and adjust whenever $\xi \leq G^{-1}(\varphi^*)$ — exactly the fixed cost rule. Under this coupling, the realised adjustment indicators coincide path by path. The two models agree on the distribution of the binary action, and a coupling exists under which they agree on the realised action as well. ■

The economic content is straightforward. In the fixed cost model, agents with low cost draws adjust and those with high draws do not. The expected payoff equals the adjustment probability times the surplus, minus the average cost paid by adjusters. In the probability choice model, the agent chooses an adjustment probability and pays a convex cost $C(\varphi)$. The theorem says these two descriptions yield exactly the same payoff and adjustment probability, provided C is constructed from G via (3).

The Fenchel conjugate structure places the two formulations on equal mathematical footing. The payoff function $\Pi^{PC}(S)$ and the cost function $C(\varphi)$ are conjugates of each other: $C^*(S) = \Pi^{PC}(S)$ and $(\Pi^{FC})^*(\varphi) = C(\varphi)$. The two express the same value function in dual variables — one in the surplus S, the other in the adjustment probability φ. This is analogous to Lagrangian duality, where primal and dual problems yield the same value at optimality. Any result proved for one variable translates immediately to the other. The two are not "different theories" but different parametrisations of a single theory. Section 3 develops a concrete illustration in which G is uniform, C is quadratic, and the equivalence can be verified by direct calculation.

Relation to discrete choice theory The Williams-Daly-Zachary (WDZ) theorem (McFadden, 1978, 1981) establishes that the expected maximum utility under random taste shocks, $E[\max_j\{u_j + \varepsilon_j\}]$, is the Fenchel conjugate of a generating function determined by the joint distribution of the shocks ε_j. For binary choice, the WDZ conjugate identity and Theorem 1 share the same algebraic skeleton (Table G.1 in Online Appendix H). The binary conjugate identity is therefore not new; the contributions of this paper lie in what it enables.

The main value of Theorem 1 is what the duality implies beyond the identity itself: (i) the mapping $G \mapsto C$ in equation (3) and the distribution-theoretic taxonomy in Propositions 4–6, which characterise exactly when and how much lumpiness matters; (ii) the structural explanation for the Khan and Thomas (2008) result and its correct scope; and (iii) the continuous parametrisation of the Calvo–Golosov-Lucas spectrum. These contributions are entirely absent from WDZ and from the discrete choice literature. The duality is the tool; the taxonomy and the pricing application are the results.

Two further distinctions are worth noting. First, the source of randomness differs: in WDZ, randomness enters payoffs via taste shocks ε_j; here, randomness enters costs via the draw $\xi \sim G$. For binary choice the two are isomorphic under a sign change. Second, the coupling is new. WDZ establishes equivalence of expected payoffs and choice probabilities; Theorem 1 further constructs a natural randomisation device under which the realised adjustment decisions coincide path by path — not merely in distribution — justifying the claim that the two formulations are literally the same model.

Relation to control costs In the game theory literature, Mattsson and Weibull (2002) and Hofbauer and Sandholm (2002) study agents who choose mixed strategies subject to convex "control costs" that penalise departures from a default action. Deviating from habitual behaviour is costly, and the cost increases as the deviation becomes more extreme. The probability choice formulation (2) has exactly this structure: $C(\varphi)$ is the cost of choosing adjustment probability φ relative to the default of not adjusting ($\varphi = 0$). Theorem 1 therefore shows that the fixed cost model *is* a control cost model. The control cost is not an ad hoc penalty imposed by the modeller. It is the endogenous cost function determined by the distribution of frictions that the agent faces.

2.3 Properties of the induced cost function

The induced cost function $C(\varphi)$ inherits its shape entirely from the distribution G. The following proposition characterises its key properties and provides the economic interpretation that underlies all subsequent applications.

Proposition 1 (Cost function properties)**.**
The induced cost function (3) *satisfies:*

(i) $C(0) = 0$, $C(1) = w\, E_G[\xi]$.

(ii) $C'(\varphi) = w\, G^{-1}(\varphi)$*: marginal cost equals the cost of the marginal adjuster.*

(iii) $C''(\varphi) = w/g(G^{-1}(\varphi)) > 0$*: strictly convex whenever* $g > 0$.

(iv) C *is* C^2 *if* g *is* C^1 *and positive.*

Proof.
Differentiating (3): $C'(\varphi) = w\, G^{-1}(\varphi) \cdot g(G^{-1}(\varphi)) \cdot [g(G^{-1}(\varphi))]^{-1} = w\, G^{-1}(\varphi)$, using the inverse function theorem. Differentiating again: $C''(\varphi) = w \cdot [g(G^{-1}(\varphi))]^{-1} > 0$. ■

Part (ii) deserves emphasis, as it carries the core economic content. The marginal cost of raising the adjustment probability by $d\varphi$ is $C'(\varphi) = wG^{-1}(\varphi)$ — the cost of the agent who is just indifferent at the current threshold. Agents are ranked by their cost draws: those with the lowest ξ adjust first. When the threshold probability is φ, the fraction φ of agents with the lowest costs have already been induced to adjust. Pushing the adjustment probability slightly higher requires extending the threshold to include agents with marginally higher cost draws. Their cost is $wG^{-1}(\varphi)$ — the φ-th quantile of the cost distribution, scaled by w. This is the economic force that makes C convex. The cheapest agents adjust first, and each subsequent increment in the adjustment probability requires inducing progressively more expensive agents to act.

Part (iii) reveals how the density g governs the curvature of C. When g is large at $G^{-1}(\varphi)$, many agents are clustered near the current threshold. A small increase in the threshold induces many additional adjusters, so the marginal cost rises slowly and C'' is low. When g is small, agents are sparse near the threshold. The same increase in adjustment probability requires a large jump in the threshold, so the marginal cost rises sharply and C'' is high. Crowded margins mean gentle curvature; sparse margins mean sharp curvature.

Corollary 1 (Shape of G determines curvature)**.**

(i) g increasing $\Rightarrow C''$ decreasing: there are relatively more agents at higher cost quantiles, so inducing additional adjustment becomes progressively easier. Adjustment is "cheap at the margin" for high φ.

(ii) g decreasing $\Rightarrow C''$ increasing: agents thin out at higher cost quantiles, so inducing additional adjustment becomes progressively more expensive. Adjustment becomes disproportionately costly at high φ.

(iii) g constant (uniform) $\Rightarrow C''$ constant: agents are equally spaced throughout the cost distribution, yielding the standard quadratic cost function with constant curvature.

Proof.
By Proposition 1(iii), $C''(\varphi) = w/g(G^{-1}(\varphi))$. Since G^{-1} is increasing, g increasing implies $g(G^{-1}(\varphi))$ is increasing in φ, so $C''(\varphi) = w/g(G^{-1}(\varphi))$ is decreasing. The argument for g decreasing and g constant is analogous. ■

The following corollary applies the mapping (3) to four canonical distributions, recovering well-known cost function specifications as special cases of the single formula $C(\varphi) = w \int_0^{\varphi} G^{-1}(u)\, du$. Each distribution represents a different assumption about the cross-sectional heterogeneity of adjustment costs, and the induced cost function inherits its functional form from the shape of G.

Corollary 2 (Canonical distributions)**.**

*(i) **Uniform** $G = U[0, \bar{\xi}]$: $C(\varphi) = \frac{w\bar{\xi}}{2}\varphi^2$. The constant density produces constant curvature, so each unit increase in the adjustment probability costs the same marginal increment. This is the standard quadratic adjustment cost used throughout the macroeconomics and industrial organisation literatures.*

*(ii) **Power law** $G(\xi) = (\xi/\bar{\xi})^{\gamma}$, $\gamma > 0$: $C(\varphi) = \frac{\gamma}{\gamma+1} w\bar{\xi}\, \varphi^{1+1/\gamma}$. The cost function is iso-elastic with elasticity $1 + 1/\gamma$, and the shape parameter γ governs the curvature profile. For $\gamma > 1$, the density $g(\xi) = \gamma\xi^{\gamma-1}/\bar{\xi}^{\gamma}$ is increasing: most cost draws are high, so cheap adjusters are scarce and C is less curved than quadratic. For $\gamma < 1$, the density is decreasing: most cost draws are low, cheap adjusters are abundant, and C is more curved than quadratic. The uniform case is $\gamma = 1$. As $\gamma \to \infty$, the density concentrates near $\bar{\xi}$ and C approaches*

linearity (the degenerate limit). As $\gamma \to 0$, the density concentrates near zero and C approaches a step function at $\varphi = 0$ (nearly free adjustment). The power law family thus spans the full range from lumpiness to frictionless adjustment, indexed by a single parameter. This family plays the central role in the pricing application of Section 4.

(iii) ***Exponential*** *$G(\xi) = 1 - e^{-\lambda\xi}$ on $[0, \infty)$: $C'(\varphi) = -\frac{w}{\lambda}\ln(1-\varphi)$, giving $C(\varphi) = \frac{w}{\lambda}[(1-\varphi)\ln(1-\varphi) + \varphi]$. The logarithmic marginal cost diverges as $\varphi \to 1$, severely penalising near-certain adjustment. This reflects the exponential distribution's unbounded support: to guarantee adjustment with probability close to one, the agent must be willing to pay arbitrarily high cost draws, which is infinitely expensive in expectation. As shown in Section 2.5, this cost function is proportional to the binary Shannon entropy, connecting the fixed cost model to the rational inattention literature (Sims, 2003; Matějka and McKay, 2015).*

(iv) ***Degenerate*** *at $\bar{\xi}$: $C(\varphi) = w\bar{\xi}\varphi$. The cost function is linear, with no strict convexity. The only optimal choices are $\varphi = 0$ (never adjust) or $\varphi = 1$ (always adjust), depending on whether S falls below or above $w\bar{\xi}$. This is the canonical binary threshold rule of the deterministic fixed cost model — the singular case analysed in Proposition 2 below.*

Proof.
Apply $C(\varphi) = w\int_0^\varphi G^{-1}(u)\,du$ from Proposition 1(ii). (i) Uniform: $G^{-1}(\varphi) = \bar{\xi}\varphi$, so $C(\varphi) = w\bar{\xi}\int_0^\varphi u\,du = \frac{w\bar{\xi}}{2}\varphi^2$. (ii) Power law: $G^{-1}(\varphi) = \bar{\xi}\,\varphi^{1/\gamma}$, so $C(\varphi) = w\bar{\xi}\int_0^\varphi u^{1/\gamma}\,du = \frac{\gamma}{\gamma+1}w\bar{\xi}\,\varphi^{1+1/\gamma}$. (iii) Exponential: $G^{-1}(\varphi) = -\lambda^{-1}\ln(1-\varphi)$, so $C(\varphi) = -\frac{w}{\lambda}\int_0^\varphi \ln(1-u)\,du = \frac{w}{\lambda}[(1-\varphi)\ln(1-\varphi)+\varphi]$. (iv) Degenerate at $\bar{\xi}$: $G^{-1}(\varphi) = \bar{\xi}$ for all $\varphi \in (0, 1]$, giving $C(\varphi) = w\bar{\xi}\varphi$. ■

The corollary illustrates the unifying power of the mapping $G \mapsto C$. The quadratic cost function, which is typically postulated as a primitive in convex adjustment cost models, arises endogenously from uniform heterogeneity. The iso-elastic family, which nests the quadratic as a special case, arises from power law heterogeneity. The Shannon entropy cost, which is the foundation of the rational inattention literature, arises from exponential heterogeneity. And the linear cost function, which is the foundation of the deterministic menu cost literature, arises from the absence of heterogeneity.

These are not four different models — they are four special cases of one model, distinguished solely by the shape of G.

2.4 Universality of the convex representation

The degenerate case — a single, known fixed cost — is the only setting in which the problem is not strictly convex. This section shows that this case is a knife-edge singularity. Any perturbation, no matter how small, restores strict convexity. The convex model is the generic case; lumpiness is the singular exception.

Proposition 2 (Universality)**.**
Let $\bar{\xi} > 0$ be a deterministic fixed cost. For any $\epsilon > 0$, perturb the cost to $\xi \sim G_\epsilon \equiv U[\bar{\xi} - \epsilon,\ \bar{\xi} + \epsilon]$. The induced cost function is

$$C_\epsilon(\varphi) = w\big(\bar{\xi} - \epsilon\big)\varphi + w\epsilon\,\varphi^2, \tag{5}$$

which is strictly convex with $C''_\epsilon(\varphi) = 2w\epsilon > 0$ for every $\epsilon > 0$. As $\epsilon \to 0$:

(i) $C_\epsilon(\varphi) \to w\bar{\xi}\,\varphi$ pointwise: the linear (degenerate) cost.

(ii) $C''_\epsilon(\varphi) \to 0$: curvature vanishes.

(iii) The optimal $\varphi^*_\epsilon(S) = \min\Big\{\frac{S - w(\bar{\xi}-\epsilon)}{2w\epsilon},\ 1\Big\}$ converges to $\mathbf{1}[S \geq w\bar{\xi}]$: the binary threshold rule.

Thus the deterministic fixed cost is the singular limit of a family of strictly convex problems.

Proof.
For $G_\epsilon = U[\bar{\xi} - \epsilon,\ \bar{\xi} + \epsilon]$, the quantile function is $G_\epsilon^{-1}(\varphi) = (\bar{\xi} - \epsilon) + 2\epsilon\varphi$ and the density is $g_\epsilon = 1/(2\epsilon)$. Applying (3):

$$C_\epsilon(\varphi) = w\int_0^\varphi G_\epsilon^{-1}(t)\,dt = w\int_0^\varphi \big[(\bar{\xi} - \epsilon) + 2\epsilon t\big]\,dt = w(\bar{\xi} - \epsilon)\varphi + w\epsilon\varphi^2.$$

Strict convexity: $C''_\epsilon(\varphi) = 2w\epsilon > 0$ for all $\epsilon > 0$. The first-order condition $S = C'_\epsilon(\varphi) = w(\bar{\xi} - \epsilon) + 2w\epsilon\varphi$ gives $\varphi^*_\epsilon(S) = [S - w(\bar{\xi} - \epsilon)]/(2w\epsilon)$, which is interior whenever S falls in the interval $(w(\bar{\xi} - \epsilon),\ w(\bar{\xi} + \epsilon))$. As $\epsilon \to 0$, this interval collapses to the single point $w\bar{\xi}$, and the interior solution collapses to a step function: φ^* jumps discontinuously from 0 to 1 at $S = w\bar{\xi}$. ■

The result overturns the standard view that convex costs are a "smooth approximation" to the "true" lumpy friction. Proposition 2 says the opposite: the lumpy model is a singular limit of the generic convex model. In the space of cost distributions, the degenerate distribution (a point mass) is measure zero. Every open neighbourhood of a point mass contains distributions with full support, and hence strictly convex induced costs. The convex formulation is the robust, generic case. The binary threshold rule is a fragile, non-generic limit.

The economic intuition is that the curvature $C''_\epsilon = 2w\epsilon$ measures the "softness" of the adjustment margin. When ϵ is large, there is substantial variation in costs. The adjustment probability responds smoothly to changes in the surplus, and the model behaves like a textbook convex cost problem. When ϵ is small, costs are nearly identical. The adjustment probability responds sharply — approximating a binary switch — and the model behaves like a threshold rule. But even an infinitesimally small ϵ produces a strictly convex problem. The lumpy limit is never actually reached for any positive ϵ.

Epistemic interpretation and structural stability The deterministic fixed cost requires the agent to know the adjustment cost with infinite precision. In practice, this never holds. A firm repricing its goods faces uncertainty about staff time, managerial attention, and logistical disruption. A plant manager cannot know the exact installation cost until the project is underway. If there is *any* uncertainty — any ϵ-band of noise — the decision problem is already smooth and convex.

This is a statement about structural stability. The deterministic fixed cost is structurally unstable: any ϵ-perturbation changes the decision rule from a discontinuous step function to a smooth, increasing function of the surplus. The convex formulation is structurally stable: small changes in G produce small changes in C and in the optimal policy. Results depending on exact degeneracy of G are knife-edge properties of a non-generic model. Results holding for all absolutely continuous G with positive density are robust.

Topological genericity The claim that lumpiness is non-generic can be made precise in the weak topology on the space of probability measures on $[0, \bar{\xi}]$. A sequence of measures μ_n converges weakly to μ if $\int f\, d\mu_n \to \int f\, d\mu$ for every bounded continuous f. The degenerate measure $\delta_{\bar{\xi}}$ (a point mass at $\bar{\xi}$) is the unique element of this space

that induces a linear cost function and a binary threshold rule. Yet every weakly open neighbourhood of $\delta_{\bar{\xi}}$ contains measures with continuous positive densities, and hence strictly convex induced costs. To see this, let U be any weakly open neighbourhood of $\delta_{\bar{\xi}}$. By the Portmanteau theorem, U contains all measures μ satisfying $|\int f_i \, d(\mu - \delta_{\bar{\xi}})| < \epsilon$ for finitely many bounded continuous test functions $f_1, \ldots, f_k$. The uniform perturbation $G_\epsilon = U[\bar{\xi} - \epsilon, \bar{\xi} + \epsilon]$ satisfies $\int f_i \, dG_\epsilon \to f_i(\bar{\xi})$ as $\epsilon \to 0$ by continuity, so $G_\epsilon \in U$ for ϵ small enough. Since G_ϵ has a continuous positive density, it induces a strictly convex C_ϵ by Proposition 2. Therefore $\delta_{\bar{\xi}}$ lies in the closure of the set of absolutely continuous measures: the degenerate (lumpy) case is a limit point of the convex (smooth) case, and every neighbourhood of it contains non-degenerate, strictly-convex instances.

Mean-preserving spreads and convexity Proposition 2 shows that any perturbation of a point mass restores strict convexity. The following result establishes the general principle: among distributions with the same mean, greater dispersion produces a lower — and more curved — cost function. The degree of convexity is monotone in the mean-preserving spread (MPS) order, the most natural partial order on distributions with a fixed mean.

Proposition 3 (Dispersion and convexity)**.**
Let G_1 and G_2 be distributions on $[0, \bar{\xi}]$ with the same mean, $E_{G_1}[\xi] = E_{G_2}[\xi]$, and let G_2 be a mean-preserving spread of G_1. Let C_1 and C_2 be the induced cost functions. Then:

(i) $C_1(1) = C_2(1) = w\,E[\xi]$*: the total cost of certain adjustment is identical.*

(ii) $C_2(\varphi) \leq C_1(\varphi)$ *for all* $\varphi \in [0, 1]$*: greater dispersion lowers the cost function in the interior.*

(iii) The marginal costs single-cross: there exists $\varphi_0 \in (0, 1)$ *such that* $C_2'(\varphi) \leq C_1'(\varphi)$ *for* $\varphi < \varphi_0$ *and* $C_2'(\varphi) \geq C_1'(\varphi)$ *for* $\varphi > \varphi_0$*.*

Consequently, greater dispersion of G lowers the cost of moderate adjustment, raises the cost of near-certain adjustment, and steepens the marginal cost schedule. The degenerate distribution (a point mass) maximises C in the interior and minimises curvature (C is linear); any spread away from it lowers C and increases curvature.

Proof.

Part (i): By Proposition 1(i), $C_i(1) = w\, E_{G_i}[\xi]$. Same mean gives $C_1(1) = C_2(1)$.

Part (ii): By Proposition 1(ii), $C_i(\varphi) = w \int_0^{\varphi} G_i^{-1}(u)\, du$. The standard quantile characterisation of mean-preserving spreads states that G_2 is a MPS of G_1 if and only if

$$\int_0^t G_2^{-1}(u)\, du \leq \int_0^t G_1^{-1}(u)\, du \quad \text{for all } t \in [0,1],$$

with equality at $t = 1$. Therefore $C_1(\varphi) - C_2(\varphi) = w \int_0^{\varphi} [G_1^{-1}(u) - G_2^{-1}(u)]\, du \geq 0$ for all φ, with equality at $\varphi \in \{0, 1\}$.

Part (iii): Since $C_i'(\varphi) = wG_i^{-1}(\varphi)$ and both quantile functions are continuous and increasing with the same integral over $[0, 1]$, while G_2^{-1} is stretched relative to G_1^{-1} (lower at the bottom, higher at the top), the two marginal cost schedules cross exactly once. ■

The economic content is intuitive. A more dispersed cost distribution means that the cheapest adjusters are cheaper and the most expensive holdouts are more expensive. The early units of adjustment probability can be achieved at lower cost (by inducing the very cheapest agents), but the last units become more expensive (the remaining agents have higher costs). The net effect is a steeper marginal cost schedule — more curvature — while the total cost of inducing everyone to adjust ($C(1) = wE[\xi]$) is unchanged.

The result connects Proposition 2 to the broader theory: Proposition 2 is the special case where $G_1 = \delta_{\bar{\xi}}$ (a point mass, giving linear C) and G_2 is any non-degenerate distribution with the same mean. Any spread away from the point mass lowers the cost function in the interior and restores curvature. Proposition 3 generalises this: the more dispersed the distribution, the lower and more curved the cost function. Dispersion of G and convexity of C are monotonically linked through the MPS order.

In the pricing application of Section 4, this linkage has direct aggregate consequences. A lower cost function means that firms with small price gaps find it cheaper to adjust, so the pool of adjusters becomes less selected — more like a random sample and less like the firms with the largest gaps. Weaker selection flattens the Phillips curve and increases monetary non-neutrality (Section 4.7). The MPS order on G thus induces a monotone ordering on the real effects of monetary shocks: holding the mean menu cost fixed, greater dispersion moves the economy toward Calvo and away from Golosov–Lucas.

2.5 Fixed costs as generalised entropies

The cost function $C(\varphi)$ measures how expensive it is to deviate from inaction. It is fully determined by G and inherits its shape from the quantile function G^{-1}. This section shows that canonical distributions G yield cost functions that coincide with well-known information-theoretic entropies, revealing that the fixed cost literature in macroeconomics, the discrete choice literature in industrial organisation, the rational inattention literature in information economics, and the sparse decision-making literature are all manifestations of the same duality viewed through different distributions G.

Proposition 4 (Entropy correspondence)**.**

(i) ***Exponential distribution****: $G(\xi) = 1 - e^{-\lambda\xi}$ on $[0, \infty)$. Then*

$$C'(\varphi) = -\frac{w}{\lambda}\ln(1-\varphi), \qquad C(\varphi) = \frac{w}{\lambda}\big[(1-\varphi)\ln(1-\varphi) + \varphi\big].$$

The adjustment probability under exponential costs is $\varphi^* = 1 - e^{-\lambda S/w}$, the CDF of the exponential distribution evaluated at S/w. Note that this is not *the logistic function $1/(1+e^{-S/\sigma})$ that arises in logit discrete choice; the two share a sigmoid shape but differ in functional form. The true logit correspondence arises in the multi-action extension (Section 2.6), where i.i.d. Gumbel-distributed costs across alternatives yield the multinomial logit.*

(ii) ***Power law distribution****: $G(\xi) = (\xi/\bar{\xi})^\gamma$ on $[0, \bar{\xi}]$. Then*

$$C(\varphi) = \frac{\gamma}{\gamma+1} w\bar{\xi}\,\varphi^{1+1/\gamma}.$$

This is proportional to φ^q with $q = 1 + 1/\gamma$, which coincides with the functional form of the Tsallis q-entropy (Tsallis, 1988), a one-parameter generalisation of Shannon entropy used in statistical mechanics. The parameter γ controls the index q: $\gamma = 1$ gives $q = 2$ (quadratic); as $\gamma \to \infty$, $q \to 1$ and the cost function approaches the Shannon form.

(iii) ***Uniform distribution****: $G = U[0, \bar{\xi}]$. Then $C(\varphi) = \frac{w\bar{\xi}}{2}\varphi^2$, a quadratic penalty. This is the simplest and most widely used convex cost specification, and it coincides with the L^2 penalty that appears in many regularisation frameworks.*

Proof.
Part (i): The quantile function is $G^{-1}(\varphi) = -\lambda^{-1}\ln(1-\varphi)$. Then

$$C(\varphi) = w\int_0^{\varphi} G^{-1}(t)\,dt = -\tfrac{w}{\lambda}\int_0^{\varphi}\ln(1-t)\,dt = \tfrac{w}{\lambda}\big[(1-\varphi)\ln(1-\varphi)+\varphi\big].$$

Parts (ii)–(iii) follow from Corollary 2. ■

Rational inattention and exponential fixed costs Sims (2003) introduces rational inattention, where agents face a Shannon mutual information cost of processing information about a continuous state. Matějka and McKay (2015) show that in the discrete-choice specialisation, this cost reduces to a Shannon entropy penalty over the choice probability vector, yielding logit choice probabilities. Proposition 4(i) shows that the cost function induced by exponentially-distributed fixed costs is exactly the Shannon entropy of the binary adjustment probability — the cost function in the binary discrete-choice rational inattention model. For binary choice, the two models are mathematically equivalent: they produce the same adjustment probabilities and the same payoffs. This equivalence does not extend to the full continuous-state RI framework of Sims (2003), where the mutual information cost has a richer structure involving Bayesian updating over a continuous state space. The precise claim is that exponential G provides a friction-based micro-foundation for the discrete-choice RI cost function, not for the full information-processing framework.

Sparsity and uniform fixed costs Gabaix (2014) penalises attention allocation with a quadratic cost, yielding "sparse" decisions that underreact to stimuli. By Proposition 4(iii), the cost function induced by uniform fixed costs is exactly this quadratic penalty. The equivalence is at the level of the attention cost function for the binary decision; the full sparsity framework of Gabaix (2014) includes additional structure (endogenous attention allocation across multiple dimensions) not captured by the binary duality. The degree of sparsity is governed by $\bar{\xi}$: a wider cost support means more costly attention and more default-biased decisions.

Scope of the correspondences The mathematical coincidences in Proposition 4 arise from the conjugate structure. The same class of convex functions appears whenever one maximises a linear objective subject to a convex penalty, regardless of what

the penalty represents economically.

The connection to rational inattention (Sims, 2003) is worth making explicit. Matějka and McKay (2015) show that when an agent faces uncertainty about a state and processes information subject to a Shannon entropy cost, the optimal choice rule takes the multinomial logit form. In the multi-action extension of Section 2.6, i.i.d. Gumbel-distributed fixed costs yield $C(\boldsymbol{\varphi}) = \sigma \sum_j \varphi_j \ln \varphi_j$ (negative Shannon entropy), and the multinomial logit emerges as the optimal mixed strategy. Every agent who faces i.i.d. extreme-value adjustment costs and uses an optimal threshold rule is behaving exactly as if they were a rational inattention agent with Shannon entropy costs. The two models make identical predictions about adjustment frequencies and payoffs.

However, the economic mechanisms are distinct. In rational inattention, the agent is uncertain about the payoff-relevant state and incurs a cost of learning. In the present framework, the agent knows the surplus S and incurs a physical cost of acting. The reduced-form $C(\varphi)$ is the same; the welfare implications are not. Rational inattention produces predictions about how agents respond to changes in the information environment (e.g., what happens when information becomes cheaper); the fixed cost model produces predictions about how agents respond to changes in the distribution of physical frictions. These are different counterfactuals. The duality identifies the mathematical structure these models share. It does not assert that the underlying economics are the same.

The distribution G as a Rosetta Stone Table 1 summarises the mapping. Five literatures — macroeconomic frictions, discrete choice, rational inattention, sparse decision-making, and control costs — are unified by a single object: the distribution G. Each literature has independently developed models that are special cases of the duality, distinguished only by the distributional assumption on costs.

Table 1: The distribution G as a Rosetta Stone

Distribution G	**Cost $C(\varphi)$**	**Form**	**Entropy**	**Literature**
Exponential	$(1-\varphi)\ln(1-\varphi)+\varphi$	Log. MC	Shannon	Rational inattention
Power law (γ)	$\varphi^{1+1/\gamma}$	Iso-elastic	Tsallis q	Generalised discrete choice
Uniform ($\gamma=1$)	φ^2	Quadratic	L^2	Sparsity
Degenerate	$\bar{\xi}\,\varphi$ on $\{0,1\}$	Linear	—	Lumpy adjustment
Type I EV (multinomial)	$\sum \varphi_j \ln \varphi_j$	Neg. entropy	Shannon (mv)	Logit discrete choice

A robustness criterion The Shannon entropy (exponential G) is the unique entropy satisfying certain axioms (additivity, continuity, maximum entropy principle). If one believes these axioms should govern the cost of deviating from default behaviour, then exponential G is the "correct" fixed cost distribution, and the logit/rational inattention model is the uniquely justified reduced form. Departures from exponential G — such as the uniform distribution used in Khan and Thomas (2008) — correspond to departures from Shannon's axioms, and can be interpreted as reflecting specific features of the adjustment technology rather than information-processing constraints.

2.6 Beyond binary choice

The duality extends naturally beyond the binary adjustment decision. This section outlines extensions to multiple actions, state-dependent surpluses, and dynamic environments. The details are straightforward applications of the conjugate structure.

Multiple actions Suppose the agent chooses among $J+1$ actions $\{a_0, a_1, \ldots, a_J\}$, where a_0 is inaction. Each a_j for $j \geq 1$ yields surplus S_j at random cost $w\xi_j$ with $\xi_j \sim G_j$, independently across actions. The agent observes all J cost draws and selects the action with the highest net payoff. The duality extends to this multi-action setting:

Proposition 5 (Multi-action duality)**.**
Let $J \geq 1$ and suppose $\xi_j \sim G_j$ independently for $j = 1, \ldots, J$, with each G_j absolutely continuous with positive density on $[0, \bar{\xi}_j]$. Define $S_0 = 0$, $\xi_0 = 0$. Then:

(i) The expected payoff $\Pi^{FC}(\mathbf{S}) = E\big[\max_{j\in\{0,\ldots,J\}}\{S_j - w\xi_j\}\big]$ equals

$$\Pi^{PC}(\mathbf{S}) = \max_{\boldsymbol{\varphi}\in\Delta^J} \sum_{j=0}^{J} \varphi_j S_j - C(\boldsymbol{\varphi}), \tag{6}$$

where $C : \Delta^J \to \mathbb{R}_+$ is jointly convex and determined by the vector of distributions $(G_1, \ldots, G_J)$.

(ii) When $\xi_j \overset{i.i.d.}{\sim}$ Gumbel(μ, σ) for all $j \geq 1$, the cost function is $C(\boldsymbol{\varphi}) = \sigma \sum_{j=0}^{J} \varphi_j \ln \varphi_j$

(negative Shannon entropy) and the optimal choice probabilities are

$$\varphi_j^* = \frac{e^{S_j/\sigma}}{\sum_{k=0}^{J} e^{S_k/\sigma}}, \tag{7}$$

the multinomial logit of McFadden (1974).

Proof.
Apply the Fenchel conjugate construction to the multi-alternative maximum and specialise to the Gumbel distribution. The detailed proof is given in Online Appendix A. ■

State-dependent surplus In most economic applications, the surplus from adjusting depends on a state variable x. In the pricing application, x is the price gap; in the investment application, x is the capital stock. The duality applies state by state: for each value of x, the threshold rule under G is equivalent to choosing $\varphi(x)$ subject to $C(\varphi)$. The resulting function $\varphi^*(x) = G(S(x)/w)$ is the *endogenous state-dependent hazard*. Its shape inherits two features: the curvature of $S(x)$, governing how the surplus varies with the state, and the shape of G, governing how the adjustment probability responds to a given surplus. The state dependence of pricing is therefore jointly determined by the profit function and the cost distribution.

Dynamics When the state x evolves stochastically over time, the value function $V(x)$ satisfies a Bellman equation where the per-period payoff uses the probability choice formulation. Since C is convex and smooth (for G with a positive density), the Bellman equation involves maximising a concave objective. Standard dynamic programming results — existence, uniqueness, concavity, smooth policy functions — apply directly. This contrasts with the fixed cost formulation, where the per-period payoff involves a non-convex max operator and the value function generically has kinks at the adjustment thresholds. The probability choice formulation simplifies the analysis not because it approximates the fixed cost model, but because it is the same model expressed in a form where convex analysis applies.

3 Application to investment

The duality's most immediate application is to investment. Fixed costs of adjustment have been a central modelling device since Doms and Dunne (1998) documented the lumpiness of plant-level investment, Cooper and Haltiwanger (2006) estimated the importance of non-convex adjustment costs, and Gourio and Kashyap (2007) explored investment spikes in general equilibrium. The seminal quantitative exercise is Khan and Thomas (2008), who ask whether micro-level lumpiness matters for aggregate investment dynamics. More recently, Oskolkov and Lippi (2026) identify the distribution of random fixed costs from Italian plant-level data and quantify the resulting capital misallocation; the distribution G that they estimate is exactly the primitive that the duality maps to a convex cost function.

Consider a firm with capital k choosing investment $I \geq 0$ subject to a random fixed cost $\xi \sim G$. If it adjusts, it pays $w\xi$ and obtains the continuation value at capital $k(1-\delta)+I$. If not, it proceeds with depreciated capital $k(1-\delta)$. The surplus from investing at the optimal scale is

$$S(k) = \max_{I \geq 0} \left\{ -I + \beta(k(1-\delta)+I)^{\alpha} \right\} - \beta(k(1-\delta))^{\alpha}, \tag{8}$$

where I^* satisfies the first-order condition $1 = \alpha\beta(k(1-\delta)+I^*)^{\alpha-1}$. Note that I^* depends on k but not on the realised cost ξ: conditional on investing at all, the optimal scale does not depend on the cost of initiating the project. This separation between the extensive margin (whether to invest) and the intensive margin (how much to invest) is a standard feature of fixed cost models, preserved exactly by the duality.

By Theorem 1, the firm's problem is exactly

$$V(k) = \max_{\varphi \in [0,1]} \left\{ \varphi \cdot S(k) - C(\varphi) \right\} + \beta(k(1-\delta))^{\alpha}, \tag{9}$$

with C given by (3). The firm chooses an investment probability φ to balance the expected benefit of investing ($\varphi \cdot S(k)$) against the cost of achieving that probability ($C(\varphi)$). If it invests, it invests at the optimal scale I^*; if not, it does nothing. The probability of investing, $\varphi^*(k) = G(S(k)/w)$, is increasing in the surplus $S(k)$. Firms further from their desired capital stock have more to gain, draw a higher threshold, and invest more frequently.

The uniform case Khan and Thomas (2008) draw fixed costs from a uniform distribution $G = U[0, \bar{\xi}]$, the simplest case of the duality. Because agents are spread evenly across the cost support, raising the adjustment threshold by a fixed amount always captures the same number of new adjusters — but each new group costs slightly more than the last. This constant rate of inclusion with linearly increasing marginal cost produces a quadratic total cost.

Formally, the quantile function is $G^{-1}(\varphi) = \bar{\xi}\varphi$ and the density is $g = 1/\bar{\xi}$. On the fixed cost side, a firm with surplus S adjusts for draws below $\xi^* = S/w$, giving adjustment probability $\varphi^* = S/(w\bar{\xi})$ and expected payoff

$$\Pi^{FC}(S) = \frac{S}{w\bar{\xi}} \cdot S - w \int_0^{S/w} \xi \cdot \frac{1}{\bar{\xi}}\, d\xi = \frac{S^2}{2w\bar{\xi}}.$$

On the probability choice side, the induced cost function is $C(\varphi) = \frac{w\bar{\xi}}{2}\varphi^2$ — a standard quadratic adjustment cost. The first-order condition of $\max_\varphi\ \varphi S - \frac{w\bar{\xi}}{2}\varphi^2$ gives $\varphi^* = S/(w\bar{\xi})$ and maximised value $S^2/(2w\bar{\xi})$: the same probability and the same payoff. The duality is exact.

The constant curvature $C'' = w\bar{\xi}$ means each firm's investment problem is a textbook convex adjustment cost problem. This provides a *partial-equilibrium* foundation for the irrelevance result of Khan and Thomas (2008). Their full result requires a general equilibrium mechanism: endogenous adjustments in the interest rate and wages absorb idiosyncratic investment spikes, so that the aggregate capital stock is smooth even if individual investment is lumpy in the fixed-cost representation. The present paper's contribution is complementary and distinct. It establishes that, under uniform costs, individual firms do not have inherently lumpy decision problems — the convex representation is exact, not an approximation obtained by aggregation. Even a single firm facing uniform fixed costs has a smooth, concave value function and a smooth policy rule in the probability choice representation.

This clarifies the logical structure of the irrelevance argument: smoothness at the individual level (established here) is a necessary condition for the Khan and Thomas (2008) result; the GE price-adjustment mechanism is an additional sufficient condition for irrelevance in aggregate dynamics. Any model that draws fixed costs from a distribution with a positive density will generate smooth individual decision problems. The law of large numbers is sufficient for smooth aggregation, but not necessary: a positive density of costs already delivers smoothness at the individual

level.

When does micro-level lumpiness matter for aggregates? The duality provides a precise, distribution-theoretic characterisation. By Proposition 1, the induced cost function $C(\varphi)$ is smooth and strictly convex if and only if G has a continuous, positive density. Smoothness fails in three specific ways:

Proposition 6 (Conditions for smoothness of C)**.**
$C \in C^2(0,1)$ with $C'' > 0$ on the open unit interval — so that $\varphi^*(S)$ is a C^1 function of the surplus — if and only if:

(i) G is absolutely continuous with density $g > 0$ on $(0, \bar{\xi})$.

(ii) g is continuously differentiable ($g \in C^1$) on $(0, \bar{\xi})$.

Both conditions are automatically satisfied by any G with a C^1, positive density. Smoothness fails when G has atoms (producing kinks in C'), when g has interior zeros (producing points where C'' diverges), or when the support of G is disconnected (producing flat regions in C').

The degenerate case — G a point mass at $\bar{\xi}$ — violates condition (i) and produces the canonical lumpy investment model. C is linear, the only optimal choices are $\varphi \in \{0, 1\}$, and the firm either invests fully or not at all.

The curvature of G and the speed of capital adjustment The mapping $G \mapsto C$ implies a precise relationship between the dispersion of fixed investment costs and the economy's aggregate investment dynamics. In the probability choice representation, the firm's optimal investment probability responds to a small change in the surplus according to

$$\frac{d\varphi^*(k)}{dS} = \frac{1}{C''(\varphi^*(k))} = \frac{g(G^{-1}(\varphi^*(k)))}{w}.$$

This is the density of cost draws at the current threshold, scaled by $1/w$. Three implications follow directly.

First, *adjustment speed is proportional to the density at the margin.* When g is high at the current threshold (many firms clustered near the adjustment margin), a small improvement in fundamentals tips many firms into investment. Aggregate

investment responds rapidly. When g is low, the same improvement tips few firms; investment responds sluggishly. The hazard rate of G at the current threshold is the structural determinant of aggregate investment dynamics, in partial equilibrium.

Second, for the uniform distribution (g constant), $d\varphi^*/dS = 1/(w\bar{\xi})$ is constant: the investment response is linear in the productivity shock for all values of the surplus. This reproduces the standard result that quadratic adjustment cost models generate a linear investment policy function.

Third, for right-skewed cost distributions (g decreasing, e.g. power law with $\gamma < 1$), $d\varphi^*/dS$ is larger for low surpluses (where few firms currently invest) and smaller for high surpluses (where many already invest). Investment responds more sharply to small shocks that move firms from inaction to action, but the response flattens for large shocks. This nonlinearity in the aggregate investment impulse response is directly governed by the shape of G and is absent from models with a uniform distribution.

Proposition 7 (Investment response and the cost density)**.**
In partial equilibrium, the sensitivity of the aggregate investment probability to a small productivity shock that raises the surplus by dS is

$$\frac{dE[\varphi^*(k)]}{dS} = E\left[\frac{g(G^{-1}(\varphi^*(k)))}{w}\right].$$

This is proportional to the expected density of cost draws at the current threshold, averaged across firms. Economies with higher g at the relevant threshold are more investment-elastic.

Dispersion as a continuous index of lumpiness The curvature of the induced cost function provides a continuous metric of "how much lumpiness matters." For the family $G = U[\mu - \epsilon, \mu + \epsilon]$, the curvature is $C'' = 2w\epsilon$, strictly increasing in the dispersion ϵ. As $\epsilon \to 0$ (holding the mean μ fixed), the curvature vanishes and the model approaches the binary threshold rule. As ϵ grows, the curvature increases and the model resembles a standard convex cost problem. The dispersion of idiosyncratic costs is thus a continuous index of the economy's position on the spectrum from fully lumpy ($\epsilon = 0$) to fully smooth (ϵ large). There is no discrete boundary between "lumpy" and "smooth" economies.

4 The menu cost–Calvo spectrum

The deepest application of the duality is to price setting, where the distinction between time-dependent (Calvo) and state-dependent (menu cost) pricing is central to New Keynesian economics. The distribution of menu costs G governs the entire pricing environment: the repricing hazard, the selection effect, the Phillips curve slope, and the degree of monetary non-neutrality.

4.1 Setup

Consider a monopolistically competitive firm with current log price p and desired log price p^*. The *price gap* is $x \equiv p^* - p$. When the firm reprices (closing the gap to zero), it gains a surplus that depends on the distance from the optimum. The standard second-order approximation of the profit function around $p = p^*$ yields

$$S(x) = Bx^2, \tag{10}$$

where $B > 0$ depends on the demand elasticity and the discount factor.[2] The quadratic form reflects the approximate symmetry of profits around the optimum. The loss from mispricing grows with the square of the price gap: a firm with a 2% gap loses roughly four times as much as one with a 1% gap.

The firm draws a menu cost $\xi \sim G$ each period and reprices if and only if the gain from repricing exceeds the cost:

$$\text{Reprice if and only if } w\xi \leq Bx^2. \tag{11}$$

4.2 The repricing hazard

By Theorem 1, the firm's repricing decision under the threshold rule is equivalent to choosing a repricing probability $\varphi(x) \in [0, 1]$ subject to the convex cost $C(\varphi)$. The first-order condition $Bx^2 = C'(\varphi) = wG^{-1}(\varphi)$ yields the optimal repricing probability as a function of the price gap:

Definition 1 (Repricing hazard)**.**
The repricing hazard — the probability that a firm with price gap x adjusts its price

[2] This is standard; see Golosov and Lucas (2007) and Alvarez et al. (2016). The quadratic loss approximation is $S(x) \approx \frac{\varepsilon}{2}x^2$ where ε is the demand elasticity, up to discounting.

— *is*

$$\Lambda(x) \equiv G\left(\frac{Bx^2}{w}\right). \tag{12}$$

This is the central object of the pricing application.[3] The repricing hazard $\Lambda(x)$ summarises everything about the firm's pricing behaviour that matters for aggregate dynamics. Conditional on (B, w) and the steady-state gap distribution f_{ss}, all aggregate pricing implications — inflation, the Phillips curve slope, the selection effect, monetary non-neutrality — flow through Λ. The distribution G is therefore the structural primitive from which aggregate pricing dynamics are derived.

Proposition 8 (Hazard properties)**.**
The repricing hazard $\Lambda(x)$ satisfies:

(i) $\Lambda(0) = G(0) = 0$*: a firm at its optimal price never reprices. The gain from doing so is zero, and every cost draw is positive.*

(ii) Λ *is symmetric in* x *and increasing in* $|x|$*: the repricing probability depends only on the magnitude of the gap, not its sign. The surplus* Bx^2 *depends only on* x^2*. Larger gaps produce larger surpluses, which exceed more cost draws.*

(iii) $\Lambda'(x) = g(Bx^2/w) \cdot 2Bx/w$*: the sensitivity depends on the density* g *evaluated at the firm's surplus. Where* g *is high, many firms are near the threshold, and a small change in the gap tips many across it. Where* g *is low, few firms are near the threshold, and the same change has little effect.*

(iv) $\Lambda(x) \to 1$ *for* $|x| \geq \sqrt{w\bar{\xi}/B}$*: when the gap is sufficiently large, the surplus exceeds even the highest cost draw, and the firm reprices with certainty. This defines the "inaction band" beyond which all firms adjust.*

4.3 Stationary distribution of price gaps

The repricing hazard $\Lambda(x)$ governs individual pricing decisions. To close the model, one must also characterise the cross-sectional density of price gaps that arises in the stationary equilibrium. Let z denote the idiosyncratic desired-price shock with

[3] The repricing hazard $\Lambda(x)$ is the "adjustment function" in the language of Caballero and Engel (1993, 1999). Those papers take the adjustment function as a primitive and characterise how its shape determines aggregate dynamics. The present contribution is to derive $\Lambda(x)$ from first principles as a function of the menu cost distribution G, establishing G as the deeper structural primitive.

density ϕ_z, and suppose that in steady state inflation is zero. The price gap evolves as follows: a firm with gap x adjusts with probability $\Lambda(x)$, resetting its gap to a fresh draw $z' \sim \phi_z$; with the complementary probability $1 - \Lambda(x)$, it does not adjust and its gap drifts to $x + z'$.

Proposition 9 (Stationary distribution)**.**
Under the repricing hazard $\Lambda(x) = G(Bx^2/w)$ and idiosyncratic shock density ϕ_z, the stationary density f_{ss} satisfies the fixed-point equation

$$f_{ss}(x) = \underbrace{\bar{\Lambda} \cdot \phi_z(x)}_{\textit{inflow from adjusters}} + \underbrace{\int (1 - \Lambda(\tilde{x})) \cdot \phi_z(x - \tilde{x}) \cdot f_{ss}(\tilde{x})\, d\tilde{x}}_{\textit{inflow from non-adjusters}}, \tag{13}$$

where $\bar{\Lambda} = \int \Lambda(x) f_{ss}(x)\, dx$ is the aggregate adjustment frequency. In steady state with zero inflation, f_{ss} is symmetric around zero.

Proof.
Consider the mass of firms arriving at gap x in the next period. Firms that adjusted in the current period have their gap reset; their new gap is determined solely by the realised shock $z' \sim \phi_z$, irrespective of their previous state. The total mass of adjusters is $\bar{\Lambda} = \int \Lambda(\tilde{x}) f_{ss}(\tilde{x})\, d\tilde{x}$, and the density of their new gaps is $\phi_z(x)$, giving the first term. Firms that did not adjust retain their current gap $\tilde{x}$ and receive a shock z', moving to $\tilde{x} + z'$. The density at x from this group is the convolution $\int (1 - \Lambda(\tilde{x})) \phi_z(x - \tilde{x}) f_{ss}(\tilde{x})\, d\tilde{x}$, giving the second term. Stationarity requires that f_{ss} is a fixed point of this mapping. Symmetry follows from the symmetry of ϕ_z and $\Lambda(x) = \Lambda(-x)$. ■

The first term captures the "reset effect": adjusting firms are drawn back toward zero, concentrating mass near the optimum. The second term captures "diffusion": non-adjusting firms drift outward as shocks accumulate. The balance between these two forces determines the shape of f_{ss}.

When Λ is high (strong state dependence, high γ), the reset effect dominates. Most firms are close to their desired prices, because the economy is effective at correcting mispricing. When Λ is low (weak state dependence, low γ), diffusion dominates. Many firms carry large price gaps for extended periods. The shape of f_{ss} matters for aggregate dynamics because it determines where firms sit when a shock arrives. A concentrated distribution means few firms have large gaps to begin with. A dispersed

distribution means the economy is far from its optimum and more exposed to nominal disturbances. The distribution G thus governs both the hazard and, through equation (13), the steady-state cross-section — closing the model.

Remark 1. *The fixed-point equation (13) corresponds, in discrete state space, to the eigenvector problem* $\mathbf{f} = \mathbf{T}'\mathbf{f}$ *where* $\mathbf{T}$ *is the transition matrix encoding the adjustment and drift probabilities. This is the object computed numerically in the quantitative implementation.*

4.4 The Calvo–Golosov-Lucas spectrum

The repricing hazard $\Lambda(x)$ encodes the firm's pricing behaviour, and its shape is determined entirely by G. The two canonical models of price rigidity in the New Keynesian literature correspond to polar extreme shapes of Λ:

Calvo (1983) In the Calvo (1983) model, each firm reprices with a fixed probability $\lambda \in (0, 1)$ regardless of its price gap. Since $\Lambda(x) = G(Bx^2/w)$ is always increasing in $|x|$, a strictly constant hazard cannot arise from any non-degenerate absolutely continuous G. The framework therefore does not literally nest Calvo; it approximates it.

The approximation requires two ingredients. First, G must be sufficiently dispersed so that $\Lambda(x)$ is nearly flat over the region where the steady-state gap distribution f_{ss} has mass. Second, the level of the hazard must be pinned down by a calibration condition. Concretely, consider a family $G_\theta(\xi) = G(\xi/\theta)$ where $\theta > 0$ scales the support and G is a fixed shape. Holding the average adjustment frequency $\bar{\Lambda} = E_{f_{ss}}[\Lambda(x)] = \lambda$ fixed, let $\theta \to \infty$. As the support of costs grows without bound, the hazard $\Lambda(x) = G(Bx^2/(w\theta)) \to 0$ for each fixed x, but the calibration condition pins θ as a function of λ and f_{ss} so that the average frequency remains λ. In this limit, $\Lambda(x)/\bar{\Lambda} \to 1$ uniformly over the support of f_{ss}: conditional on the calibrated average frequency, the hazard becomes state-independent. This is the Calvo approximation.

Note that the power law $\gamma \to 0$ limit studied below is distinct: with fixed $\bar{\xi}$, it corresponds to near-certain adjustment ($\Lambda(x) \to 1$ for $x \neq 0$), not to a constant interior hazard. The two limits differ in whether the scale $\bar{\xi}$ is held fixed (cheap adjustment limit) or adjusted with γ to hold $\bar{\Lambda}$ fixed (Calvo approximation). This distinction matters for calibration: in quantitative applications, one should use the

frequency-calibrated approximation, not the $\gamma \to 0$ limit.

Golosov and Lucas (2007): step-function hazard Golosov and Lucas (2007) study a model with a single, known menu cost (G degenerate at $\bar{\xi}$). The repricing hazard is a step function: $\Lambda(x) = \mathbf{1}[Bx^2 \geq w\bar{\xi}]$. The firm adjusts with probability one if the gap is large enough, and zero otherwise. There is no smooth interior. This produces the canonical Ss rule with inaction band $(-\bar{x}, \bar{x})$ where $\bar{x} = \sqrt{w\bar{\xi}/B}$.

The step-function hazard generates maximal selection: adjusters are precisely the firms with the largest price gaps, and no firm with a small gap ever adjusts. Each repricing event does maximum work in moving the aggregate price level, because only the most mispriced firms reprice. This selection dramatically reduces the real effects of monetary shocks relative to Calvo.

The power law family The power law $G(\xi) = (\xi/\bar{\xi})^\gamma$, $\gamma > 0$, provides a tractable one-parameter family that spans the entire spectrum between Calvo and Golosov-Lucas:

$$\Lambda(x) = \left(\frac{Bx^2}{w\bar{\xi}}\right)^\gamma, \qquad C(\varphi) = \frac{\gamma}{\gamma+1} w\bar{\xi}\, \varphi^{1+1/\gamma}. \tag{14}$$

This family nests the uniform distribution ($\gamma = 1$), approximates Golosov-Lucas ($\gamma \to \infty$), and extends to right-skewed cost distributions ($\gamma < 1$).

Example 1 (Power law menu costs)**.**
For $G(\xi) = (\xi/\bar{\xi})^\gamma$:

(i) $G^{-1}(\varphi) = \bar{\xi}\varphi^{1/\gamma}$.

(ii) $C(\varphi) = \frac{\gamma}{\gamma+1} w\bar{\xi}\, \varphi^{(\gamma+1)/\gamma}$.

(iii) $C'(\varphi) = w\bar{\xi}\, \varphi^{1/\gamma}, \quad C''(\varphi) = \frac{w\bar{\xi}}{\gamma}\varphi^{(1-\gamma)/\gamma}$.

(iv) $\Lambda(x) = \left(\frac{Bx^2}{w\bar{\xi}}\right)^\gamma$.

The parameter γ governs the entire spectrum. As γ increases, the hazard $\Lambda(x)$ becomes steeper — responding more sharply to changes in the price gap — and the pricing environment becomes more state-dependent. The cases are:

- $\gamma = 1$: Uniform distribution. Quadratic cost. Linear hazard in x^2. Moderate selection. This is the environment studied by Dotsey et al. (1999) and Khan and Thomas (2008).

- $\gamma \to \infty$: Degenerate distribution. The hazard approaches a step function, with firms below the threshold barely adjusting and those above it adjusting almost certainly. Maximum selection. This is the Golosov and Lucas (2007) model.
- $\gamma < 1$: Right-skewed costs (many small costs, few large). The density $g(\xi) = \gamma\xi^{\gamma-1}/\bar{\xi}^{\gamma}$ is decreasing, meaning most firms face small costs but a few face very large ones. The hazard is *concave* in x^2: firms with moderate gaps adjust frequently, but the marginal increase in adjustment for very large gaps is small. Adjustment is widespread but poorly targeted — much of the economy's repricing effort goes toward firms with small gaps that contribute little to closing the aggregate price gap. Selection is *weaker than under uniform costs*, and non-neutrality is correspondingly larger.
- $\gamma \to 0$: In the limit, $\Lambda(x) \to \mathbf{1}[x \neq 0]$: every firm with any nonzero gap adjusts. The cost function is approximately linear: $C(\varphi) \approx w\bar{\xi}\varphi$, meaning adjusting is cheap and nearly all firms reprice every period. This is a "cheap adjustment" limit distinct from Calvo: the adjustment rate is high because costs are low, not because the hazard is state-independent.

4.5 The selection effect

The *selection effect* is the central force identified by Golosov and Lucas (2007) as the reason why state-dependent pricing generates much smaller real effects of monetary shocks than Calvo pricing. When repricing is state-dependent, firms that adjust tend to have the largest price gaps. They make large price changes, contributing disproportionately to aggregate price adjustment. The price level therefore responds more quickly to nominal shocks than if adjusters were drawn randomly (as in Calvo), leaving less room for real effects.

The duality framework provides a precise characterisation of this force.

Definition 2 (Selection intensity)**.**
Given a cross-sectional density $f(x)$ of price gaps, the average absolute adjusted price change *is*

$$\overline{|x|}^{adj} = \frac{\int |x| \cdot \Lambda(x) \cdot f(x)\, dx}{\int \Lambda(x) \cdot f(x)\, dx} = \frac{E[\Lambda(x) \cdot |x|]}{E[\Lambda(x)]}. \tag{15}$$

This is the mean absolute price change among repricing firms. The selection intensity *is the proportional excess of the average adjuster's absolute price change over the*

population average:

$$\mathcal{S} \equiv \frac{\overline{|x|}^{adj}}{E[|x|]} - 1. \tag{16}$$

The use of $|x|$ rather than x is essential. In the zero-inflation steady state, f_{ss} is symmetric, so $E[x] = 0$ and a ratio involving $E[x]$ would be undefined. The absolute value captures the economically relevant quantity: how large are the price changes made by adjusters relative to the population?

When Λ is constant (Calvo), $\mathcal{S} = 0$: repricing firms are a random draw from the population, so their average absolute price change equals the population average. Repricing is untargeted. Firms with tiny gaps are just as likely to adjust as those with large gaps. Much of the adjustment effort is wasted on firms that barely need it. This is why Calvo produces maximum non-neutrality: the price level moves slowly despite frequent repricing. When Λ is a step function (Golosov-Lucas), only firms beyond the threshold adjust, and $\mathcal{S}$ is maximal. Every repricing event involves a large price change, so the price level catches up quickly. In between, the degree of selection is governed by how sharply Λ increases with $|x|$.

Proposition 10 (Selection and the menu cost distribution)**.**
Fix the cross-sectional density f. The selection intensity $\mathcal{S}$ is:

(i) Increasing in the responsiveness of Λ to $|x|$, i.e., in $\Lambda'(x)/\Lambda(x)$. The more sharply the hazard increases with the gap, the more adjusters are concentrated among high-gap firms.

(ii) Higher when g is smaller (agents sparse near the threshold), since a sparse density means Λ is more responsive to x^2. When g is large, Λ is nearly flat and selection is weak.

(iii) Zero in the Calvo limit (constant Λ) and maximal in the Golosov-Lucas limit (step Λ).

Proof.
Decompose the adjustment-weighted mean as $\overline{|x|}^{adj} = E[|x|] + \mathrm{Cov}(\Lambda(x)/\bar{\Lambda},\ |x|)$. The covariance is positive when Λ is increasing in $|x|$, and its magnitude is governed by the elasticity $\partial \ln \Lambda / \partial \ln |x| = 2 \cdot g(Bx^2/w)/G(Bx^2/w) \cdot Bx^2/w$, which is twice the hazard rate of G at the firm's surplus. The full derivation is in Online Appendix E. ■

4.6 Aggregate inflation and the Phillips curve

The repricing hazard $\Lambda(x)$ governs individual pricing behaviour. Aggregate inflation aggregates this across the cross-section. Let $f_t(x)$ denote the cross-sectional density of price gaps at time t. Aggregate inflation is

$$\pi_t = E_{f_t}[\Lambda(x_t) \cdot x_t] = \int \Lambda(x)\, x\, f_t(x)\, dx. \tag{17}$$

Each firm contributes in proportion to $\Lambda(x) \cdot x$. Firms with large gaps and high repricing probabilities contribute the most; firms near the optimum contribute almost nothing.

4.6.1 State-dependent New Keynesian Phillips curve

To understand how aggregate inflation responds to nominal disturbances, consider a small aggregate shock μ_t that shifts the desired price of all firms by μ_t, moving the cross-sectional distribution of gaps from $f_{ss}(x)$ (the zero-inflation steady state) to $f_{ss}(x - \mu_t)$.

Proposition 11 (Nonlinear Phillips curve)**.**
Under the quadratic surplus approximation $S(x) = Bx^2$ and menu cost distribution G, aggregate inflation satisfies

$$\pi_t = E_t\left[G\left(\frac{Bx_t^2}{w}\right) \cdot x_t\right]. \tag{18}$$

For small aggregate disturbances around a zero-inflation steady state with gap distribution f_{ss}:

$$\pi_t \approx \kappa(G) \cdot \mu_t, \tag{19}$$

where the distribution-dependent slope *is*

$$\kappa(G) = \underbrace{\int G\left(\frac{Bx^2}{w}\right) f_{ss}(x)\, dx}_{\textit{extensive margin (frequency)}} + \underbrace{\int g\left(\frac{Bx^2}{w}\right) \frac{2Bx^2}{w}\, f_{ss}(x)\, dx}_{\textit{intensive margin (selection)}}. \tag{20}$$

Proof.
Write $f_t(x) = f_{ss}(x - \mu_t)$ and differentiate $\pi_t = \int G(Bx^2/w)\, x\, f_{ss}(x - \mu_t)\, dx$ with respect to μ_t at $\mu_t = 0$. Integrating by parts and expanding $\frac{d}{dx}[G(Bx^2/w) \cdot x]$ yields the

two terms: the first from G itself (extensive margin) and the second from $g \cdot 2Bx^2/w$ (intensive margin). The full derivation is in Online Appendix B. ■

The two terms in (20) decompose the Phillips curve slope into economically distinct components. The extensive margin (first term) captures the average repricing frequency: holding fixed which firms adjust, a nominal shock shifts all gaps by μ_t, and repricing firms adjust their prices by this amount. This is the only margin under Calvo, where Λ is constant.

The intensive margin (second term) captures the selection effect: a nominal shock changes the surpluses of firms near the threshold, tipping some into adjustment and others out. The firms tipped in have large gaps and make large price changes, amplifying the price response. The intensive margin is absent under Calvo ($g = 0$ almost everywhere in the limit) and dominant under Golosov-Lucas (the degenerate G concentrates all adjustment at the threshold).

4.6.2 Comparative statics on the slope

Corollary 3 (How G governs the Phillips curve)**.**

(i) ***Calvo limit****: When G is highly dispersed, g is small across the region where f_{ss} has mass, and the intensive margin is negligible. The slope approaches $\kappa \approx \bar{\Lambda}$, approximating the textbook New Keynesian Phillips Curve.*

(ii) ***Golosov-Lucas****: When G is degenerate, $G(Bx^2/w) = \mathbf{1}[|x| \geq \bar{x}]$ and $g = \delta_{\bar{\xi}}$. The intensive margin dominates: inflation is driven by selection of firms at the threshold. The slope is very steep, implying low non-neutrality.*

(iii) ***Uniform****: When $G = U[0, \bar{\xi}]$, the slope simplifies to*

$$\kappa^U = \frac{3B}{w\bar{\xi}} Var_{ss}(x). \tag{21}$$

The slope depends on the steady-state dispersion of price gaps. Economies with more dispersed gaps have steeper Phillips curves, because the selection margin is stronger when more firms have large gaps near the threshold.

(iv) ***Power law*** γ*: The slope* κ *is strictly increasing in* γ*. Higher* γ *means more concentrated menu costs, a steeper hazard, stronger selection, and a steeper Phillips curve.*

4.7 Non-neutrality decomposition

The degree of monetary non-neutrality is inversely related to the price flexibility index κ. When κ is high, prices adjust quickly and real effects are small. When κ is low, prices are sticky and real effects are large. The duality provides a clean decomposition of κ into two forces.

Proposition 12 (Frequency vs. selection)**.**
The real effect of a monetary shock dm *is proportional to* $(1-\kappa)\,dm$*. The flexibility index decomposes as*

$$\kappa(G) = \underbrace{\bar{\Lambda}}_{\textit{frequency}} + \underbrace{(\kappa - \bar{\Lambda})}_{\textit{selection wedge}}, \tag{22}$$

where $\bar{\Lambda} = E_{f_{ss}}[\Lambda(x)]$ *is the average adjustment frequency. Non-neutrality is governed by the* selection wedge $\kappa - \bar{\Lambda}$*.*

The frequency component $\bar{\Lambda}$ is the average repricing rate. Under Calvo, $\kappa = \bar{\Lambda} = \lambda$ and this is the only component. The selection wedge $\kappa - \bar{\Lambda}$ captures the additional flexibility from the fact that repricing firms are disproportionately those with the largest gaps. It is zero under Calvo and positive whenever Λ is increasing in $|x|$.

A key insight follows. Two economies can share the same average repricing frequency yet differ dramatically in non-neutrality. Frequency measures how many firms reprice. Selection measures how well those firms are targeted.

Consider an economy where 24% of firms reprice each quarter. If repricing is concentrated among the most mispriced firms, selection is strong and real effects are small. If repricing is spread evenly across all firms, selection is weak and real effects are large. The difference is entirely attributable to the selection wedge, which depends on the shape of G.

Corollary 4 (Ordering of non-neutrality)**.**
Across models calibrated to the same average frequency $\bar{\Lambda}$*:*

(i) ***Calvo benchmark****: selection wedge* $= 0$ *(since* Λ *is constant). Maximum non-neutrality. Repricing is random. Firms with tiny gaps are just as likely to*

adjust as firms with large gaps. The price level moves slowly despite frequent adjustment. All flexibility comes from frequency; selection contributes nothing. Within the present framework, this is approximated when G is highly dispersed.

(ii) ***Any absolutely continuous*** *G: positive selection wedge. Intermediate non-neutrality. Adjusters are disproportionately high-gap firms. The average price change among adjusters exceeds the population average. This amplifies the aggregate price response beyond what frequency alone would predict.*

(iii) ***Golosov-Lucas****: maximum selection wedge. Minimum non-neutrality. Only firms beyond the inaction band reprice. They are the most mispriced firms in the economy. Every price change is large and moves the price level toward its new equilibrium. Little room remains for real effects.*

For the power law family, non-neutrality is strictly decreasing in γ.

This recovers the well-known result of Golosov and Lucas (2007) that selection reduces non-neutrality, but embeds it in a continuous framework parametrised by G. Rather than comparing two discrete models and noting that one produces more non-neutrality, the duality provides a continuous mapping from G to non-neutrality. The two canonical models are polar limiting cases.

Proposition 13 (Monotonicity in γ)**.**
For the power law family $G(\xi) = (\xi/\bar{\xi})^\gamma$ with fixed $\bar{\xi}$ and B/w, calibrated to the same average adjustment frequency:

(i) *Selection intensity $\mathcal{S}$ is strictly increasing in γ.*

(ii) *Monetary non-neutrality is strictly decreasing in γ.*

(iii) *The Phillips curve slope at $\mu = 0$ is strictly increasing in γ.*

Thus γ indexes "how Calvo-like" the pricing environment is. Low γ (dispersed costs, weak selection, large non-neutrality) resembles Calvo. High γ (concentrated costs, strong selection, small non-neutrality) resembles Golosov-Lucas. The distribution G determines the economy's position on this spectrum.

Proof.
By Corollary 5 below, for the power law family $\kappa = (1+2\gamma)\bar{\Lambda}$ and the selection wedge equals $2\gamma\bar{\Lambda}$, both holding for any symmetric f_{ss} and any calibrated $\bar{\Lambda} > 0$.

(i) The selection intensity is $\mathcal{S} = (\kappa - \bar{\Lambda})/\bar{\Lambda} = 2\gamma$, strictly increasing in γ.

(ii) Monetary non-neutrality is proportional to $1 - \kappa = 1 - (1 + 2\gamma)\bar{\Lambda}$, strictly decreasing in γ for any $\bar{\Lambda} > 0$.

(iii) The Phillips curve slope is $\kappa = (1 + 2\gamma)\bar{\Lambda}$, strictly increasing in γ.

In each case, calibrating two power law economies to the same $\bar{\Lambda}$ (by adjusting $\bar{\xi}$) preserves γ as the sole determinant of κ, $\mathcal{S}$, and non-neutrality. ■

Remark 2. *Proposition 13 is stated within the power law family, which provides a clean one-parameter ordering at equal frequency. The underlying mechanism — concentrated G steepens the adjustment hazard, amplifying selection — is not family-specific. A general monotonicity result would require a stochastic-order condition on G that ensures higher $\kappa/\bar{\Lambda}$ whenever G is "more concentrated" in a suitable sense. We leave this extension for future work.*

The following proposition makes the frequency–non-neutrality independence precise.

Proposition 14 (Frequency does not determine non-neutrality)**.**
Fix f_{ss} and B/w. There exist absolutely continuous distributions G_1, G_2 with:

(i) The same average repricing frequency: $\bar{\Lambda}_1 = E_{f_{ss}}[\Lambda_1(x)] = E_{f_{ss}}[\Lambda_2(x)] = \bar{\Lambda}_2$.

(ii) Different flexibility indices: $\kappa(G_1) \neq \kappa(G_2)$, hence different selection wedges and different real effects of nominal shocks.

Proof.
By (20), $\bar{\Lambda}$ depends on $E[G(Bx^2/w)]$ while κ depends additionally on $E[g(Bx^2/w) \cdot 2Bx^2/w]$. Matching the first does not determine the second. For the power law $G(\xi) = (\xi/\bar{\xi})^\gamma$, direct computation gives $g(Bx^2/w) \cdot 2Bx^2/w = 2\gamma\,\Lambda(x)$, so $\kappa = (1 + 2\gamma)\bar{\Lambda}$. Two power laws with different γ, calibrated to the same $\bar{\Lambda}$ by adjusting $\bar{\xi}$, produce different κ. ■

Corollary 5 (Selection-frequency ratio for power laws)**.**
For $G(\xi) = (\xi/\bar{\xi})^\gamma$:

$$\kappa = (1 + 2\gamma)\,\bar{\Lambda}, \qquad \textit{selection wedge} = 2\gamma\,\bar{\Lambda}. \tag{23}$$

The ratio $\kappa/\bar{\Lambda} = 1 + 2\gamma$ depends only on γ, not on f_{ss}, B, or w.

Table 2 illustrates. Six distributions calibrated to the same repricing frequency $\bar{\Lambda} = 0.10$ produce selection wedges ranging from 0 to 0.60. The implied cumulative output effects differ by a factor of three.

Table 2: Same repricing frequency, different selection wedges

Distribution G	$\bar{\Lambda}$	κ	$\kappa - \bar{\Lambda}$	$1-\kappa$
Calvo	0.10	0.10	0	0.90
Power law ($\gamma = \frac{1}{2}$)	0.10	0.20	0.10	0.80
Exponential	0.10	0.27	0.17	0.73
Uniform ($\gamma = 1$)	0.10	0.30	0.20	0.70
Power law ($\gamma = 2$)	0.10	0.50	0.40	0.50
Power law ($\gamma = 3$)	0.10	0.70	0.60	0.30

Notes: Each row is calibrated to $\bar{\Lambda} = 0.10$. The last column, $1-\kappa$, is proportional to the cumulative real output effect of a monetary shock. *Power law entries:* $\kappa = (1+2\gamma)\bar{\Lambda}$, which holds for any symmetric f_{ss}. *Exponential entry:* $G(\xi) = 1 - e^{-\lambda\xi}$ with $f_{ss} = N(0, \sigma_x^2)$. Setting $\rho = B\sigma_x^2/w$, the χ^2 moment generating function gives $\bar{\Lambda} = 1 - (1 + 2\lambda\rho)^{-1/2}$ and intensive margin $\lambda\rho(1 + 2\lambda\rho)^{-3/2}$. Calibrating $\lambda\rho$ to match $\bar{\Lambda} = 0.10$ gives $\kappa \approx 0.27$.

Figure 1 complements the table with the dynamic counterpart. Four power law economies — from near-Calvo ($\gamma = 0.20$) through uniform ($\gamma = 1$) to near-Golosov-Lucas ($\gamma = 5$) — are calibrated to the same quarterly repricing frequency. Each economy is hit with a one-time 1% monetary shock. The normalised output gap $y(t)/\delta$ is tracked as prices catch up to the new money supply.

The mechanism is selection. When γ is low, repricing is nearly random. Firms with small and large gaps adjust at similar rates. The price level drifts slowly toward its new equilibrium, and the real output effect is large and persistent. When γ is high, repricing is sharply targeted at the most mispriced firms. Each adjustment event moves the price level by a large amount. The aggregate gap closes quickly and the real output effect is small and short-lived. The dashed line is the pure Calvo benchmark (exponential decay at rate $\bar{\Lambda}$), representing the zero-selection limit. All four state-dependent economies converge faster than Calvo, and the gap widens with γ.

4.8 Relation to Alvarez, Le Bihan, and Lippi (2016)

Alvarez et al. (2016) derive a sufficient-statistic result for monetary non-neutrality. In

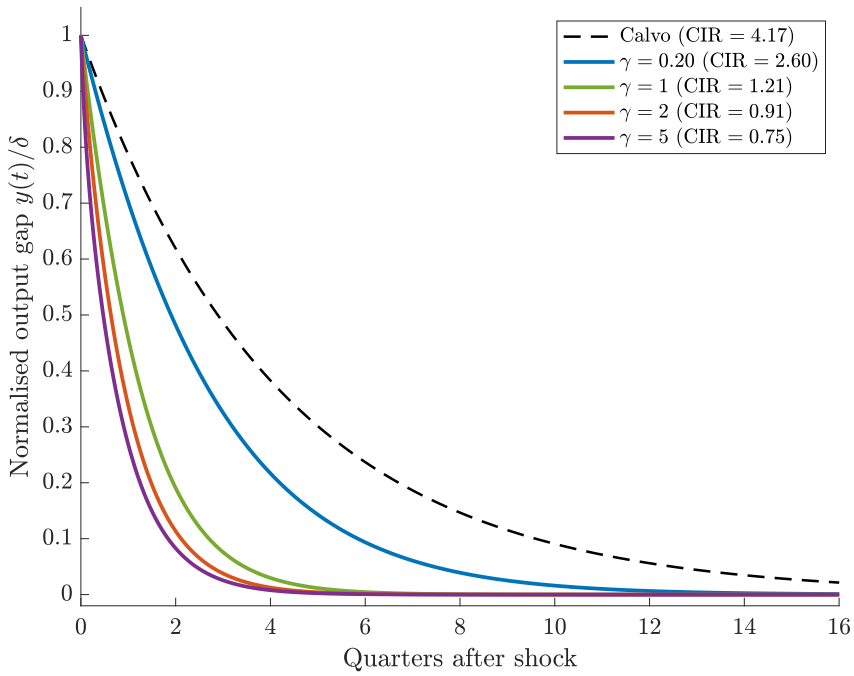

Figure 1: Output impulse responses to a 1% monetary shock. All economies are calibrated to the same average repricing frequency $\bar{\Lambda} = 0.24$ quarterly; they differ only in the power law parameter γ of the menu cost distribution $G(\xi) = (\xi/\bar{\xi})^{\gamma}$. Higher γ means stronger selection — adjusting firms have larger price gaps — so the price level catches up faster and the cumulative real output effect (CIR) is smaller. The Calvo benchmark (dashed) has zero selection and the largest CIR.

a class of menu cost models, the cumulative impulse response of output to a monetary shock is proportional to the ratio of the kurtosis of price changes to their frequency:

$$\text{CIR} = \frac{1}{6}\frac{\text{Kurtosis of price changes}}{\text{Frequency of price changes}}. \tag{24}$$

This characterises non-neutrality in terms of two moments directly observable in micro price data, without requiring knowledge of the underlying model structure.

The present framework provides a complementary characterisation. When G is the primitive, both kurtosis and frequency are endogenous functions of G:

$$\text{Frequency} = E_{f_{ss}}[\Lambda(x)] = E_{f_{ss}}\left[G\left(\frac{Bx^2}{w}\right)\right], \tag{25}$$

and kurtosis depends on how $\Lambda(x)$ reweights the gap distribution $f_{ss}(x)$.

The connection between kurtosis and selection is direct. When selection is weak, firms adjust regardless of gap size. Some adjusters have small gaps and make small price changes. Others have large gaps and make large ones. This mix of small and large price changes drives kurtosis up. When selection is strong, only the most mispriced firms adjust. They all have similarly large gaps. Price changes are uniformly large with little variation in size, so kurtosis is low. In the Golosov-Lucas limit, all price changes are the same size and kurtosis reaches its minimum of one.

The two approaches are complementary. Alvarez et al. (2016) characterise non-neutrality in terms of *observable* moments — kurtosis and frequency — providing a model-free diagnostic applicable directly to micro price data. The present paper characterises it in terms of the *structural* friction — the distribution G — from which those moments are derived. G determines the sufficient statistics, and the sufficient statistics summarise the aggregate implications of G. The practical value of G is that it reveals the mechanism behind the moments: why kurtosis is high, not just that it is.

4.9 Summary of the spectrum

Table 3 summarises how the menu cost distribution G maps into the key objects of the pricing application.

Table 3: The state-dependence spectrum

	Calvo	**Uniform G**	**Golosov-Lucas**
Distribution G	Dispersed	$U[0, \bar{\xi}]$	Point mass at $\bar{\xi}$
$C(\varphi)$	Highly curved	Quadratic	Linear on $\{0, 1\}$
$C''(\varphi)$	$\to \infty$	$w\bar{\xi}$	0 or ∞
Hazard $\Lambda(x)$	$\approx$ constant	$Bx^2/(w\bar{\xi})$	Step function
Selection	$\approx$ none	Moderate	Maximum
NKPC slope	$\approx$ constant	State-dependent	Very steep
Non-neutrality	Maximum	Intermediate	Minimum
Power law γ	$\to 0^+$	1	$\to \infty$

5 Further applications

The duality applies to any extensive-margin decision with heterogeneous fixed costs. This section sketches further applications to illustrate the framework's generality.

Entry and exit Consider a market with a pool of potential entrants, each drawing an entry cost $\xi \sim G$. A potential entrant enters if and only if expected discounted profit exceeds $w\xi$. By the duality, aggregate entry behaves as if a representative entrant chooses an entry probability φ subject to $C(\varphi)$. The free-entry condition is $S = C'(\varphi) = wG^{-1}(\varphi)$: the marginal entrant's cost equals the marginal cost of raising the entry rate. This micro-founds the smoothness of aggregate entry rates and the use of convex entry cost functions in reduced-form models of industry dynamics.

Search and matching In labour markets, vacancy creation may involve heterogeneous fixed costs — recruiting expenses, facility preparation, regulatory compliance — that differ across firms or time. When these costs are drawn from G, the duality implies that aggregate vacancy creation behaves as if firms face a convex posting cost. This micro-founds the convex vacancy-posting cost standard in Pissarides (2000). The convexity parameter reflects the shape of the cost distribution across firms, not a technological curvature in the posting process.

Human capital investment Workers deciding whether to acquire education face heterogeneous costs — tuition, foregone earnings, psychic costs of effort — that vary across individuals. If these costs are distributed according to G, the aggregate education rate is governed by a convex cost function determined by G. The supply elasticity is $1/C''(\varphi) = g(G^{-1}(\varphi))/w$: the density of workers at the margin, scaled by the wage. When g is high at the threshold, a small increase in the skill premium induces many new workers to acquire education. When g is low, the same increase induces few.

Technology adoption Heterogeneous adoption costs generate a smooth adoption S-curve if G has a positive density. The equivalence reframes the Bass (1969) diffusion model as a probability choice problem, where the convex cost captures the increasing difficulty of inducing adoption among higher-cost agents.

Default and financial frictions In models of credit markets with heterogeneous borrowers, default occurs when a borrower's idiosyncratic cost of repayment exceeds the benefit of maintaining creditworthiness (Bernanke et al., 1999). The duality implies that the aggregate default rate is a smooth function of fundamentals with curvature governed by the cost distribution. The sensitivity of the default rate to fundamentals depends on the density of borrowers near the default threshold, just as the repricing hazard's sensitivity depends on the density of firms near the adjustment threshold.

Labour supply and participation Heterogeneous fixed costs of labour force participation — childcare, commuting, health costs — generate a smooth participation rate as a function of wages. The convex cost formulation provides a structural foundation for the labour supply elasticity: $1/C''(\varphi) = g(G^{-1}(\varphi))/w$, the density of workers at the margin. This micro-founds the convexification in Rogerson (1988): with indivisible labour, lotteries over employment convert the non-convex individual problem into a convex aggregate problem. The duality shows that lotteries are unnecessary — heterogeneous participation costs achieve the same convexification structurally. The distribution G of participation costs is the primitive that determines the aggregate labour supply elasticity; its shape governs whether participation responds sharply or gradually to changes in wages.

6 Discussion

Interpretation and scope The duality is a mathematical identity: the expected payoff under the threshold rule equals the Fenchel conjugate of C, and the two formulations produce the same observable outcomes at every level of aggregation. What appears as binary, lumpy behaviour in the fixed cost representation appears as smooth, probability-weighted behaviour in the convex representation. Micro-level "lumpiness" is an artefact of the extensive-margin representation; conversely, any convex adjustment cost model has an equivalent fixed-cost micro-foundation via the distribution G.

The identity does not, however, assert that the economic mechanisms behind those cost functions are the same across contexts. A firm facing random physical costs and an agent processing information subject to Shannon entropy constraints may share the

same reduced-form $C(\varphi)$, but they face different decision problems and have different welfare implications. The duality identifies the mathematical structure these models have in common. The economic content resides in the interpretation of G and the source of the friction.

Micro jumps, macro humps The generic smoothness of aggregate dynamics despite micro-level lumpiness resonates with a broader pattern in macroeconomics. Auclert et al. (2020) document that individual households make discrete, lumpy consumption adjustments ("micro jumps"), yet aggregate impulse responses to monetary policy are smooth and hump-shaped ("macro humps").

The duality provides a structural explanation. Whenever the cost distribution G has a positive density — the generic case by Proposition 2 — binary micro-level decisions aggregate into smooth macroeconomic dynamics. Micro lumpiness survives to the aggregate level only in the knife-edge of a degenerate G. Macro humps are the expected outcome of micro jumps. Cost heterogeneity is the structural mechanism that smooths extensive-margin decisions into gradual aggregate responses. The same logic extends beyond pricing to any setting with heterogeneous fixed costs, including the labour supply and investment margins central to Auclert et al. (2020).

What the duality does not imply The equivalence established in Theorem 1 is an identity of value functions and adjustment probabilities. It does not imply equivalence of welfare, economic interpretation, or policy conclusions across the literatures sharing the same cost function form.

Welfare. In the fixed cost formulation, welfare depends on the realised cost draws $w\xi$ paid by adjusting agents. The expected adjustment cost is $\int_0^{G^{-1}(\varphi^*)} w\xi \, dG(\xi) = C(\varphi^*)$, the cost function evaluated at the optimal adjustment probability. In the probability choice formulation, $C(\varphi)$ is a reduced-form penalty with no direct welfare interpretation unless the underlying G is known. Comparing welfare across two economies with the same $C(\varphi)$ but different G requires knowledge of G, not just C.

Identification. The two representations are observationally equivalent at every level of aggregation. Micro data on individual adjustment decisions, cross-sectional gap distributions, and aggregate dynamics are all consistent with both formulations. The identifiable object is the shape of G (equivalently, C), not the "type" of friction. A researcher estimating a convex adjustment cost model and a researcher estimating

a menu cost distribution from the same micro data are, under the duality, estimating the same object.

Causality and mechanism. The mathematical equivalence does not imply that the causal mechanisms are the same. An economy where firms literally draw random menu costs each period (Dotsey et al., 1999) and one where firms choose repricing probabilities subject to a convex penalty make different predictions about how the cost distribution changes with inflation and how firms respond to cost-reducing innovations. The duality is a statement about reduced-form outcomes given G; it is silent on the deeper question of where G comes from and how it responds to policy.

Empirical implications The two formulations are observationally equivalent. Micro data alone cannot distinguish heterogeneous fixed costs from convex probability costs. What the data *can* identify is the shape of G.

Estimating G from micro data — from the cross-sectional distribution of price changes or the adjustment hazard — is the empirically informative exercise. Different shapes of G produce different hazard functions, different distributions of price changes, and different aggregate dynamics. Whether one estimates a menu cost distribution or a convex adjustment cost, the underlying object is the same: G and C are related by the bijection (3).

Oskolkov and Lippi (2026) provide a concrete illustration. They recover G from the size distribution and frequency of plant-level investments in Italian data. The duality implies that their estimated G simultaneously determines the convex cost function C, the investment hazard, and the resulting capital misallocation — all through the single mapping in equation (3). The entropy correspondence further suggests that the shape of G encodes information about the nature of the friction. Exponential G is consistent with information-processing costs; bounded-support G with physical adjustment costs.

7 Conclusion

Heterogeneous fixed costs and convex adjustment costs are Fenchel conjugate duals — equivalent exactly, not approximately. The deterministic fixed cost is a knife-edge; any perturbation restores strict convexity, making lumpiness the singular exception. The distribution G emerges as the fundamental primitive. It determines the cost function,

the repricing hazard, the selection effect, the Phillips curve slope, and monetary non-neutrality. It nests Calvo and Golosov–Lucas as polar cases. Through the entropy correspondence, it unifies five literatures — macroeconomic frictions, discrete choice, rational inattention, sparse decision-making, and control costs — as manifestations of one duality under different distributional assumptions. The structural primitive is the distribution G; all aggregate implications — the hazard, the selection effect, the Phillips curve, and non-neutrality — are determined by it.

References

Alvarez, F., H. Le Bihan, and F. Lippi (2016): "The Real Effects of Monetary Shocks in Sticky Price Models: A Sufficient Statistic Approach," *American Economic Review*, 106, 2817–2851.

Auclert, A., M. Rognlie, and L. Straub (2020): "Micro Jumps, Macro Humps: Monetary Policy and Business Cycles in an Estimated HANK Model," Working Paper 26647, National Bureau of Economic Research.

Bass, F. M. (1969): "A New Product Growth for Model Consumer Durables," *Management Science*, 15, 215–227.

Bernanke, B. S., M. Gertler, and S. Gilchrist (1999): "The Financial Accelerator in a Quantitative Business Cycle Framework," in *Handbook of Macroeconomics*, Elsevier, vol. 1, 1341–1393.

Caballero, R. J. and E. M. R. A. Engel (1993): "Microeconomic Rigidities and Aggregate Price Dynamics," *European Economic Review*, 37, 697–711.

——— (1999): "Explaining Investment Dynamics in U.S. Manufacturing: A Generalized (S, s) Approach," *Econometrica*, 67, 783–826.

——— (2007): "Price Stickiness in Ss Models: New Interpretations of Old Results," *Journal of Monetary Economics*, 54, 100–121.

Calvo, G. A. (1983): "Staggered Prices in a Utility-Maximizing Framework," *Journal of Monetary Economics*, 12, 383–398.

Caplin, A. S. and D. F. Spulber (1987): "Menu Costs and the Neutrality of Money," *Quarterly Journal of Economics*, 102, 703–725.

Cooper, R. W. and J. C. Haltiwanger (2006): "On the Nature of Capital Adjustment Costs," *Review of Economic Studies*, 73, 611–633.

Doms, M. and T. Dunne (1998): "Capital Adjustment Patterns in Manufacturing Plants," *Review of Economic Dynamics*, 1, 409–429.

Dotsey, M., R. G. King, and A. L. Wolman (1999): "State-Dependent Pricing and the General Equilibrium Dynamics of Money and Output," *Quarterly Journal of Economics*, 114, 655–690.

Gabaix, X. (2014): "A Sparsity-Based Model of Bounded Rationality," *Quarterly Journal of Economics*, 129, 1661–1710.

Golosov, M. and R. E. Lucas, Jr. (2007): "Menu Costs and Phillips Curves," *Journal of Political Economy*, 115, 171–199.

Gourio, F. and A. K. Kashyap (2007): "Investment Spikes: New Facts and a General Equilibrium Exploration," *Journal of Monetary Economics*, 54, 1–22.

Hofbauer, J. and W. H. Sandholm (2002): "On the Global Convergence of Stochastic Fictitious Play," *Econometrica*, 70, 2265–2294.

Khan, A. and J. K. Thomas (2008): "Idiosyncratic Shocks and the Role of Nonconvexities in Plant and Aggregate Investment Dynamics," *Econometrica*, 76, 395–436.

Mattsson, L.-G. and J. W. Weibull (2002): "Probabilistic Choice and Procedurally Bounded Rationality," *Games and Economic Behavior*, 41, 61–78.

Matějka, F. and A. McKay (2015): "Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model," *American Economic Review*, 105, 272–298.

McFadden, D. (1974): "Conditional Logit Analysis of Qualitative Choice Behavior," in *Frontiers in Econometrics*, ed. by P. Zarembka, Academic Press, 105–142.

——— (1978): "Modelling the Choice of Residential Location," in *Spatial Interaction Theory and Planning Models*, ed. by A. Karlqvist et al., North-Holland, 75–96.

——— (1981): "Econometric Models of Probabilistic Choice," in *Structural Analysis of Discrete Data*, ed. by C. F. Manski and D. McFadden, MIT Press, 198–272.

Midrigan, V. (2011): "Menu Costs, Multiproduct Firms, and Aggregate Fluctuations," *Econometrica*, 79, 1139–1180.

Nakamura, E. and J. Steinsson (2010): "Monetary Non-Neutrality in a Multisector Menu Cost Model," *Quarterly Journal of Economics*, 125, 961–1013.

Oskolkov, A. and F. Lippi (2026): "Misallocation with Lumpy Investment," Discussion Paper 21212, Centre for Economic Policy Research.

Pissarides, C. A. (2000): *Equilibrium Unemployment Theory*, MIT Press, 2nd ed.

Rogerson, R. (1988): "Indivisible Labor, Lotteries and Equilibrium," *Journal of Monetary Economics*, 21, 3–16.

Sims, C. A. (2003): "Implications of Rational Inattention," *Journal of Monetary Economics*, 50, 665–690.

Tsallis, C. (1988): "Possible Generalization of Boltzmann-Gibbs Statistics," *Journal of Statistical Physics*, 52, 479–487.

Vavra, J. (2014): "Inflation Dynamics and Time-Varying Volatility: New Evidence and an Ss Interpretation," *Quarterly Journal of Economics*, 129, 215–258.